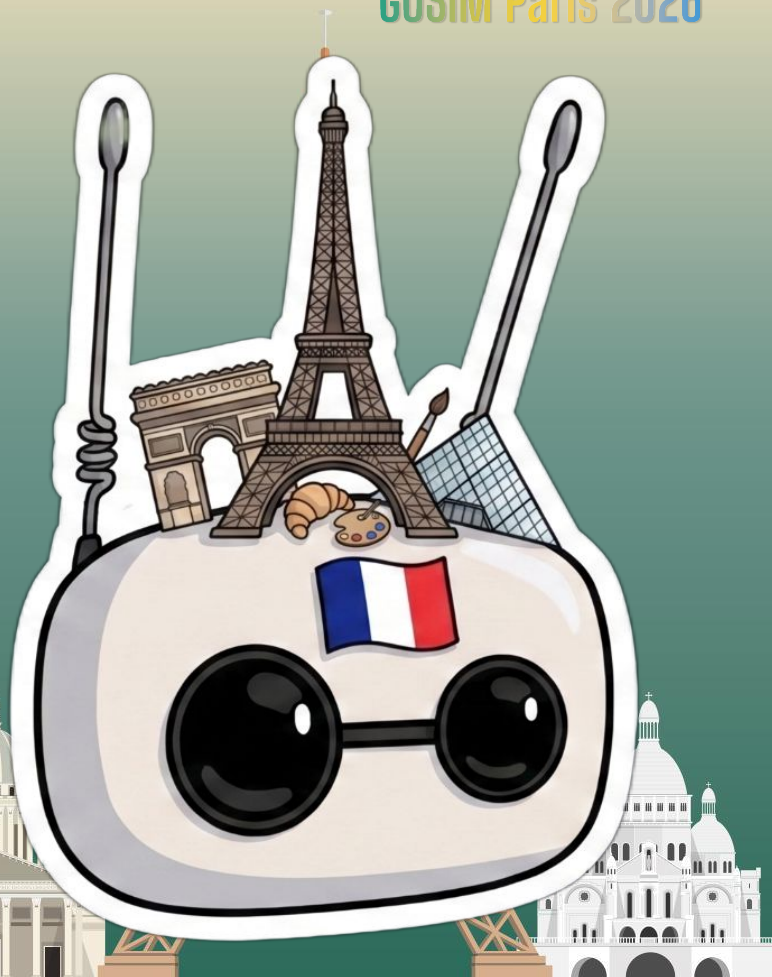


GOSIM

GOSIM Paris 2026

GOSIM Paris 2026

May 5-6, 2026 Station F, Paris



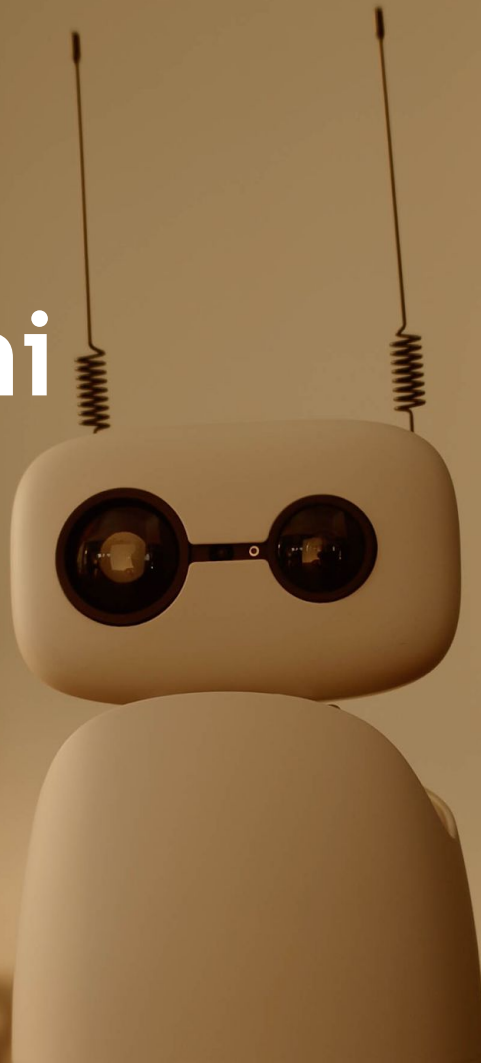
Embodied AI

Reachy Mini

How a tiny robot captured me

Tom Mulder - Pollen Robotics

Robotics + AI Builder
Reachy Mini Community Manager



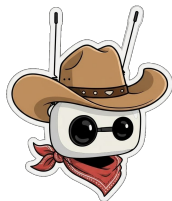
Camera + 4-Mic Array 12MP wide-angle vision, directional audio via XMOS chip, it can see and hear you

Onboard compute with a Raspberry Pi CM4 with WiFi and battery, or local compute connected your own machine via USB

Stewart platform with 6 DoF head, 360° base, expressive antennas, 9 motors that make it feel alive

Full control over movement, camera, audio. Apps via Hugging Face. JS coming really soon





Instantly Captured

Saw Reachy Mini announced by Pollen Robotics & Hugging Face. An open-source robot you build yourself, program in Python, and teach new things? I had to be part of this.



Beta Tester

Asked to join the beta program. Got in. Immediately started building, testing, breaking things, and feeding everything back to the team at Pollen Robotics.



Community Builder

What started as testing turned into something bigger. Now doing the majority of community support helping new builders on Discord, filing issues, shaping what comes next.



Building Your Own Robot

Reachy Mini ships in parts.
That's not a limitation, it's how people learn.



Building Your Own Robot

Why It Matters



- You understand what's inside
- You can fix it when something breaks
- You can expand and modify
- You feel ownership and pride
- You learn mechanics, electronics, and software at once

The Build

2-3 hrs

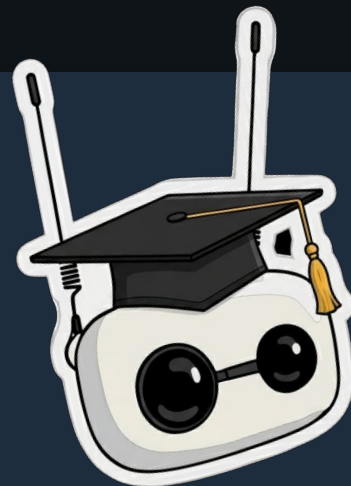
average assembly time

<45 min

fastest

2 weeks

with remote support ;)



Not just a device you unpack and turn on, something you built with your own hands.



Instant curiosity in Action

My son's school • 4–6 year olds • No instructions needed

"Why does a male deer have antlers?"

"Can you talk like a farmer from a local province with an accent?"

"What does a cow eat?"



Who It Inspires



Parents, Students & Kids

A first real encounter with robotics, programming, and AI. Not a toy, tool a platform they grow into. Assembly teaches patience and understanding of systems.



Developers & Hackers

Full Python SDK, open hardware designs, Hugging Face integration. Prototype AI behaviors, test vision models, build conversational agents on a real robot.



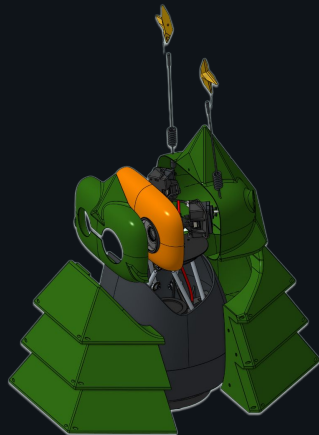
Educators & Researchers

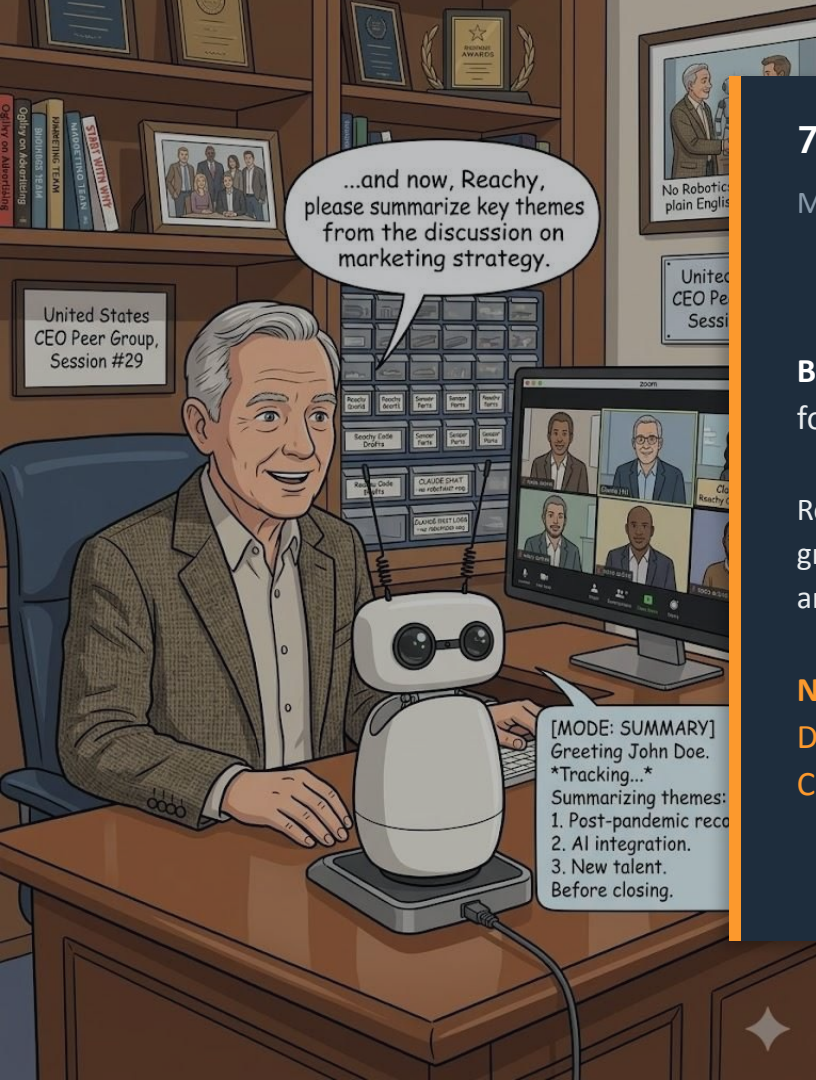
Affordable enough for classrooms. Open enough for labs. Covers mechanics, electronics, software, and AI in one kit. Simulation available for teaching without hardware.



Makers & Artists

Customizable shell, hackable hardware, unused ports waiting to be discovered. A canvas for creative expression, from digital art to interactive installations.





78 yo, Friendly gentlemen from the United States

Marketing executive. Runs CEO peer groups.

Built a voice-controlled AI co-facilitator
for live CEO peer group sessions on Zoom.

Reachy has a full personality, four facilitation modes, greets 29 members by name, tracks the conversation, and summarizes key themes before closing.

No SDK. No robotics background.

Described what he needed in plain English.

Claude wrote the code.

From 4-year-olds asking about deer antlers to a 78-year-old running AI-facilitated CEO sessions.

The Community

What happens when you make robotics open and accessible



>7K

robots shipped

~ 10K

Orders



Active Discord



How It Works

People help each other build and troubleshoot.
Share apps and behaviors on Hugging Face.
File issues, suggest features, shape the roadmap.
From first-time coders to experienced engineers.



What People Build

Conversational companions with LLMs.
Vision-based gesture recognition.
Educational tools for classrooms.
Creative art installations.
Health and wellbeing applications.

github.com/pollen-robotics/reachy_mini - huggingface.co/docs/reachy_mini

Why Open Source Matters



Reachy Mini isn't just open source by license. It's built that way on purpose.



Understand, don't just consume

When you can read the manual, trace the wiring, and modify the SDK, you're learning, not just using.



Fix and improve it yourself

Something broken? Or missing? You have the files, the manual, and a community that's been there before.



Learn from each other

Apps shared on Hugging Face. Hardware mods on GitHub. Every project someone shares teaches the next person.

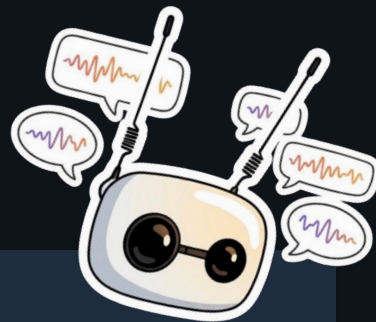


Lower the barrier, raise the ceiling

Affordable enough for a student. Deep enough for a researcher. One platform, many journeys.

What's Possible

Three things I built to show the range of what you can do



Controllable RGB Eyes

Found an unused USB port. Connected an ESP32 with two multi-color LEDs. Now Reachy has expressive, color-changing eyes that react to its mood. Hardware hacking made easy because you know what's inside.



Conversational AI + Virtual Dress-Up

Tool calling via the conversation app: Reachy takes your photo, sends it to an open-source vision model on a DGX Spark, and dresses you virtually. Robotics + LLMs + vision in one flow.



60 GHz mmWave Health Sensing

Added a 60 GHz mmWave sensor. Reachy detects micro-movements, heart rate, breathing patterns. Opens up possibilities for health monitoring, stress detection, elderly care.

Controllable RGB Eyes

Making use of unused ports

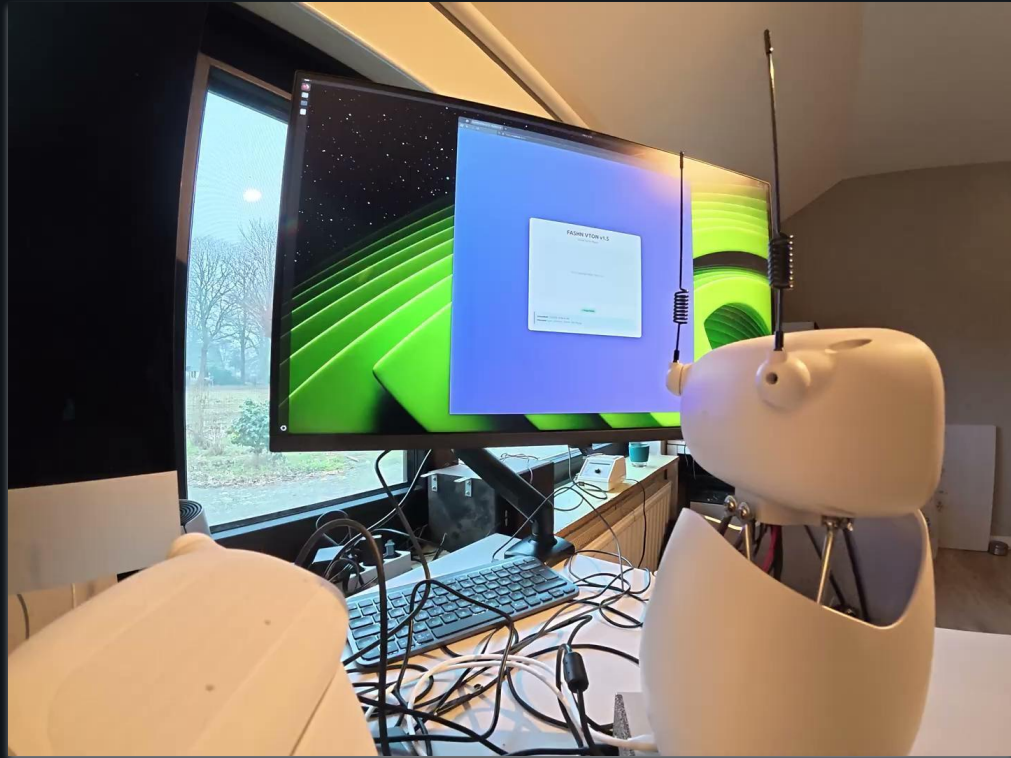


Found an unused USB port in the Lite.
Connected an ESP32 with two multi-color
LEDs. Now Reachy has expressive,
color-changing eyes that react to its mood.
Hardware hacking made easy.



Virtual-Try-On

Linking to other models (Local and remote)



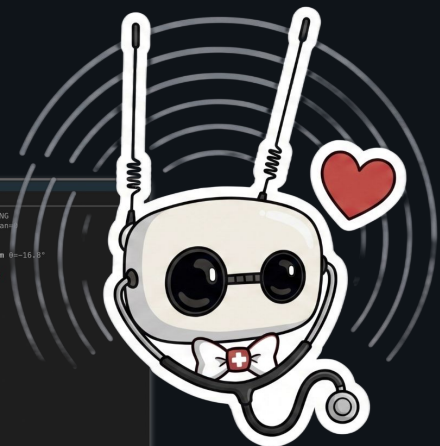
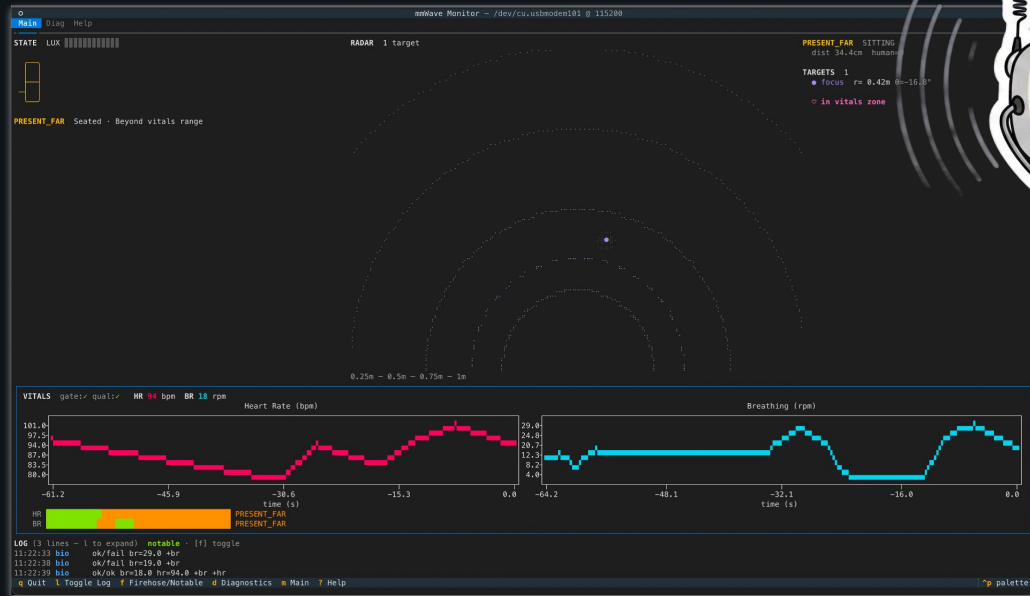
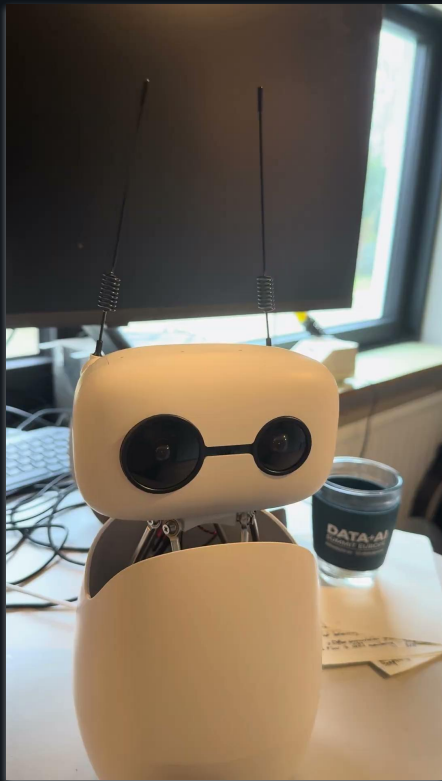
Tool calling via the conversation app: Reachy takes your photo, sends it to an open-source vision model on a local DGX Spark, and dresses you virtually, using an open source VTON model.

Robotics + LLMs + vision + generative models in one flow.



Heart rate & Breathing

Sensing a person's state



Added a 60 GHz mmWave sensor, connected to an ESP32 as the interface.

Reachy detects people's micro-movements, heart rate & breathing patterns.

Opens up possibilities for health monitoring, stress detection, elderly care.

REACHY MINI LITE SYSTEM ARCHITECTURE

INTELLIGENT PERCEPTION • EMPATHETIC CONVERSATION • PRIVACY FIRST

60 GHz mmWave SENSOR



60 GHz
mmWave
sensor



Detects up
to 6 people



Tracks person
locations



Measures
heart rate +
breathing



Up to 1.5 m
for one person
at a time



ESP32-S6 MICROCONTROLLER



Custom firmware
on ESP32-S6

Custom serial
protocol via
USB-C →



MACBOOK PRO M4

SOFTWARE STACK



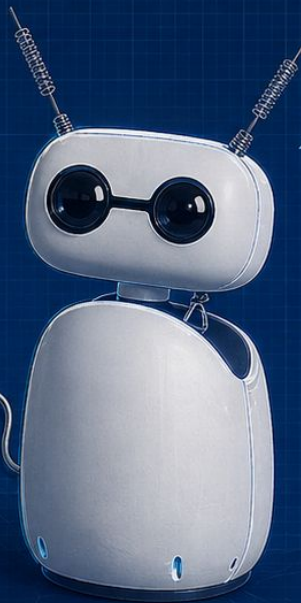
Reachy Mini Daemon
Device communication & state management



Reachy Mini Conversation App
Conversational interface & context



LLM tool call: `read_sensor_status()`
Get real-time sensor insight via tool call



You seem stressed.
How are you doing?



Health and
stress monitoring



Asks how you
are doing, based
on actual sensor
insight



Extends perception
beyond human
sight and sound



~€20 hardware
sensor as an
LLM tool

mmWave SENSOR FIELD OF VIEW (TOP VIEW)



SELECTED PERSON (WITHIN 1.5 m)



mmWave
Sensor



ESP32-S6
Controller



USB-C



MacBook Pro M4
(Host)



Reachy Mini Lite
(Robot)

KEY CAPABILITIES

- Real-time multi-person detection (up to 6)
- 3D position tracking with mm-level precision
- Physiological signals: heart rate, breathing rate
- Works in low light and privacy-friendly (no camera)
- Provides rich human context to LLMs as a tool
- Enables empathetic, insightful conversations

NOTES

All processing of sensor data can remain local for privacy. Designed for developers, researchers, and creators building the next generation of empathetic AI.





Thanks!

